



KEESLER AIR FORCE BASE BILOXI, MS
REPLACE MAGNOLIA DFAC AHUS, B7409
PROJECT NUMBER: MAHG241006
ISSUE DATE: APRIL 2026



KEESLER AIR FORCE BASE BILOXI, MISSISSIPPI
 REPLACE MAGNOLIA DFAC AHJUS. B7409
 PROJECT NO.: MAHG24-1006

COVER SHEET

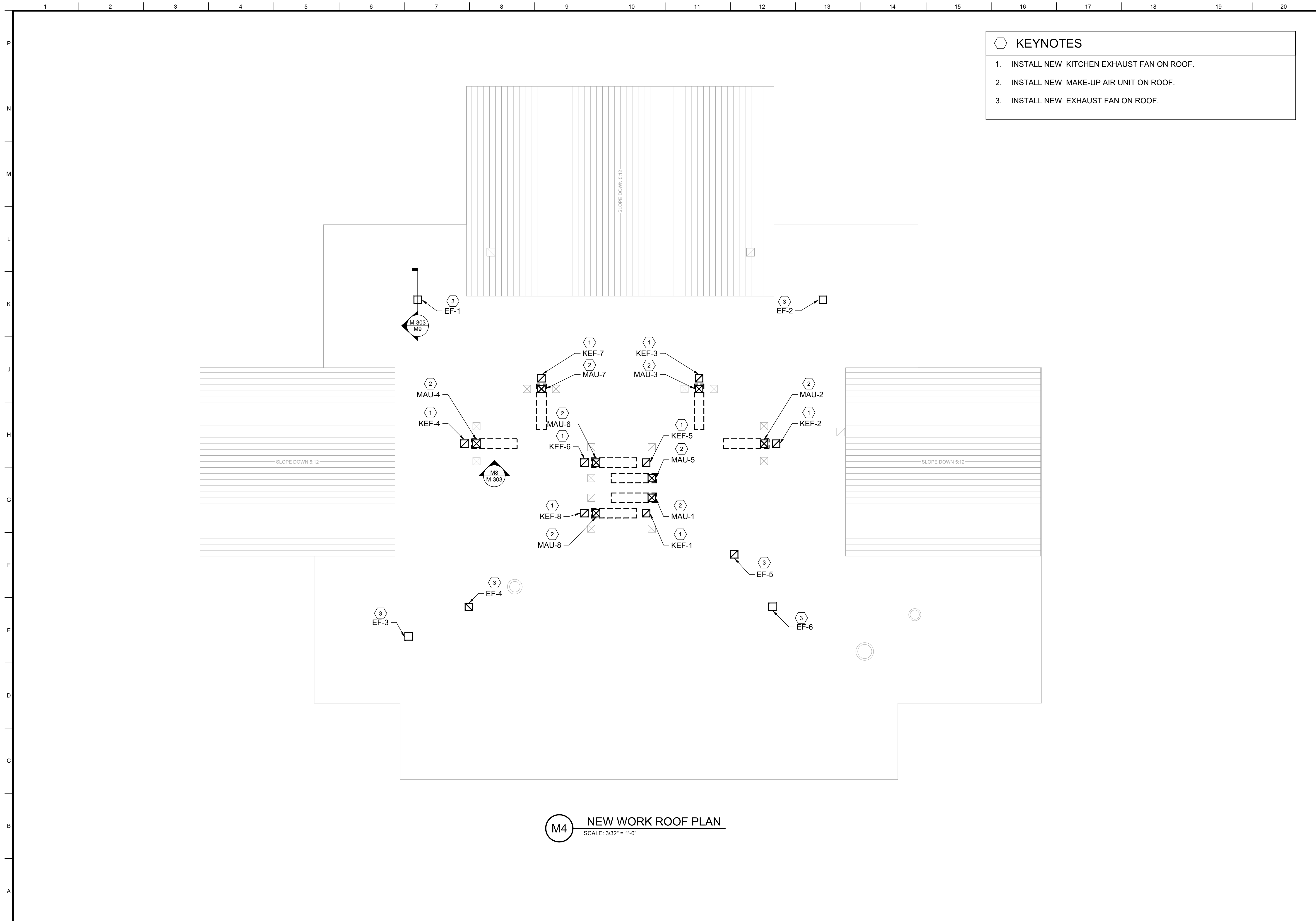
SHEET ID

G-001

81ST CIVIL ENGINEER SQUADRON
 508 L STREET B4705
 BILOXI, MS 39534

DESIGNED BY:	ISSUE DATE:	MARK	DESCRIPTION	DATE
A. TOWE	April 26			
DRAWN BY:	SOLICITATION NO.:			
M. TANNER				
CHECKED BY:	CONTRACT NO.:			
A. TOWE				
SUBMITTED BY:				
J. CLEVELAND				
SIZE:				
ANSI D				

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[illegible]

81ST CIVIL ENGINEER SQUADRON 501 1 STREET B4705 BILOXI, MS 39534	DESIGNED BY:		ISSUE DATE:
	A. TOWE		April 26
	DRAWN BY:		SOLICITATION NO.:
	M. TANNER		
	CHECKED BY:		CONTRACT NO.:
	SUBMITTED BY:		
	J. CLEVELAND		
	SIZE:		
	ANSI D		

KEESLER AIR FORCE BASE BIOL. MISSISSIPPI
REPLACE MAGNOLIA DFAC AHUS, B7409
PROJECT NO.: MAHG241006

SHEET ID
M-201

FOR REFERENCE ONLY

CFM PER ZONE

ZONE	AHU NO.			CFM
	1	2	3	
1	X			2060
2	X			2060
3	X			1740
4	X			480
5	X			1860
AHU 1 TOTAL CFM				8200
1		X		1300
2		X		1165
3		X		500
4		X		1785
5		X		550
AHU 2 TOTAL CFM				5300
1			X	1955
2			X	525
3			X	1830
4			X	2080
5			X	2080
AHU 3 TOTAL CFM				8470

LEGEND

V.D. MANUAL VOLUME BALANCING DAMPER



ISSUE DATE: PHASE 1
DESIGNED BY: #####
DRAWN BY: M. TANNER
CHECKED BY: J. CLEVELAND
SUBMITTED BY: J. CLEVELAND
SIZE: ANSI D

81ST CIVIL ENGINEER SQUADRON
508 L STREET B4705
BILLOXI, MS 39534

KEESLER AIR FORCE BASE BILLOXI, MISSISSIPPI
REPLACE MAGNOLIA DFAC AHUS, B7409
PROJECT NUMBER: MAHG241006

SHEET ID
M-203

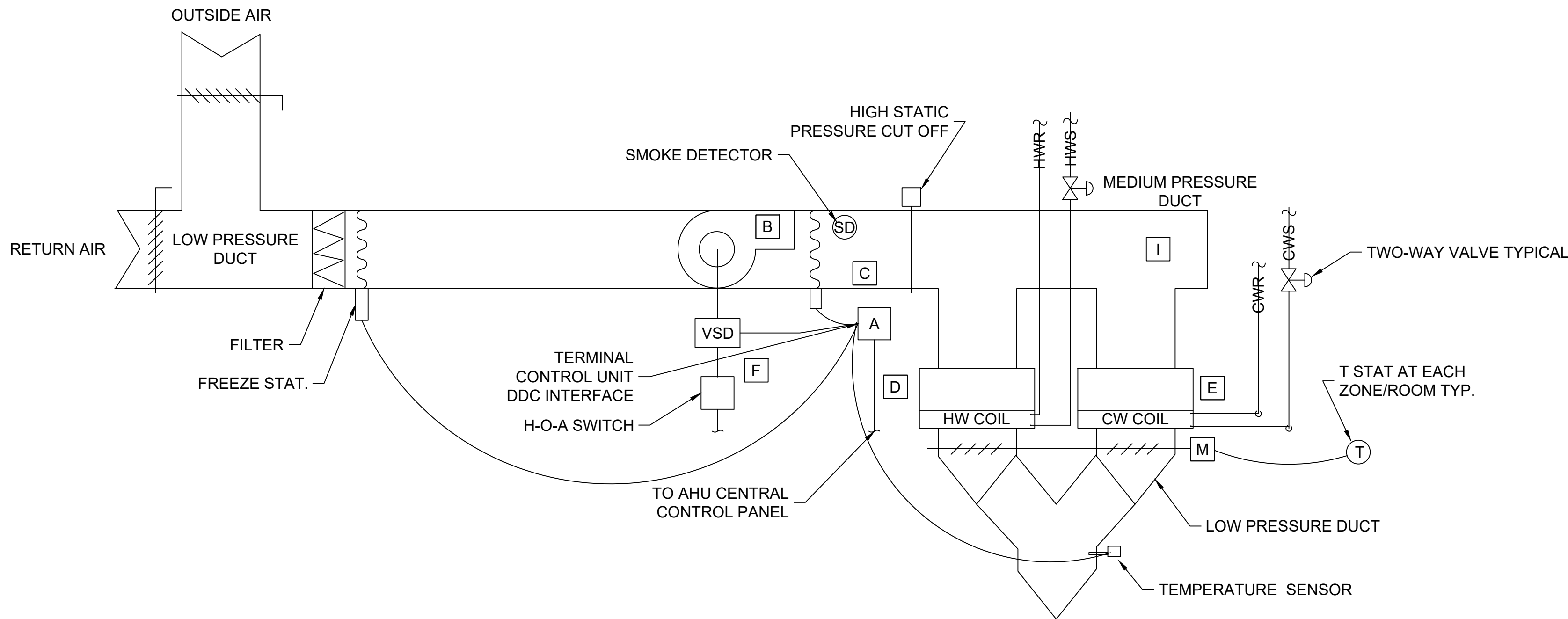
NEW WORK PLAN

100% DESIGN SUBMITTAL

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
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SEQUENCE OF OPERATION NOTES - UNITS SHALL HAVE A CONSTANT VOLUME OF OUTSIDE AIR. THE FANS SHALL HAVE VARIABLE SPEED DRIVES. THE FAN SPEED SHALL BE MODULATED TO MAINTAIN THE SET POINT PRESSURE IN THE SUPPLY DUCT. SUPPLY DUCT. PRESSURE SHALL BE SENSED BY A PRESSURE TRANSMITTER LOCATED ABOVE THE CEILING. DIFFERENTIAL PRESSURE ACROSS THE FAN SHALL BE USED TO MONITOR FAN STATUS. - COIL MULTI-ZONE SHALL BE PROVIDED BY THE HEATING COIL IN THE AIR HANDLING UNIT. ALL HEATING COILS SHALL HAVE 2-WAY MODULATING VALVES. THE CONTROL VALVES SHALL MODULATE THE FLOW OF HOT WATER THROUGH THE COILS TO MAINTAIN EACH ROOM OR ZONE SET POINT TEMPERATURE. 70 DEGREES F OF THE THERMOSTAT. - COIL MULTI-ZONE SHALL HAVE A TEMPERATURE SENSOR LOCATED DOWNSTREAM OF THE FAN. WHEN THE SUPPLY AIR TEMPERATURE RISES ABOVE THE SET POINT, A 2-WAY, CHILLED WATER CONTROL VALVE IS SIGNALLED TO MODULATE OPEN. THE 2-WAY VALVE SHALL MODULATE THE FLOW OF CHILLED WATER THROUGH THE COIL TO MAINTAIN THE SET POINT TEMPERATURE. INITIAL SET POINT SHALL BE 55 DEGREES F. - EACH ROOM/ZONE THERMOSTAT SHALL BE ADJUSTABLE BY THE OCCUPANT TO WITHIN A 2 DEGREE RANGE OF THE DDC SYSTEM SET POINT. THE SET POINT, AND THE SIZE OF THE LIMITED RANGE SHALL BE ADJUSTABLE AT THE DDC CENTRAL CONTROL PANEL. - EACH AIR HANDLING UNIT SHALL BE CONTROLLED BY A DDC CONTROL PANEL THROUGH A LOCAL CONTROL UNIT. EACH CONTROL PANEL SHALL BE PROGRAMMED WITH THE SCHEDULES AND SET POINTS OF EACH UNIT. - THE SUPPLY FANS SHALL BE ENERGIZED CONTINUOUSLY DURING THE OCCUPIED HOURS. - UPON DETECTION OF 35 DEGREES F MIXED AIR TEMP, AS SENSED BY THE	FREEZE PROTECTION TEMP SENSOR, THE FREEZE PROTECTION SEQUENCE SHALL BE ACTIVATED. THE AIR HANDLER FAN AND TEMP CONTROLS SHALL BE DEACTIVATED. THE CHILLED AND HOT WATER VALVES SHALL BE FULLY OPENED. THE FREEZE PROTECTION SEQUENCE SHALL REMAIN ACTIVATED UNTIL THE MIXED AIR TEMP RISES ABOVE 40 DEG F. - UPON DETECTION OF SMOKE THROUGH THE FIRE ALARM SYSTEM, THE AIR HANDLER FAN SHALL BE SHUT DOWN BY HARDWIRE INTERLOCK. - UPON DETECTION OF HIGH WATER LEVEL IN THE CONDENSATE PAN, THE AIR HANDLER FAN SHALL BE SHUT DOWN. - EXHAUST FANS, KITCHEN EXHAUST FANS AND MAKE-UP AIR UNITS SERVING AN INDIVIDUAL HOOD SHALL BE INTERLOCKED TO OPERATE SIMULTANEOUSLY. FANS OPERATE FROM SWITCH AT HOOD. PROVIDE STATUS ON FANS.
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M7 HVAC CONTROLS
SCALE: 1' = 1'-0"

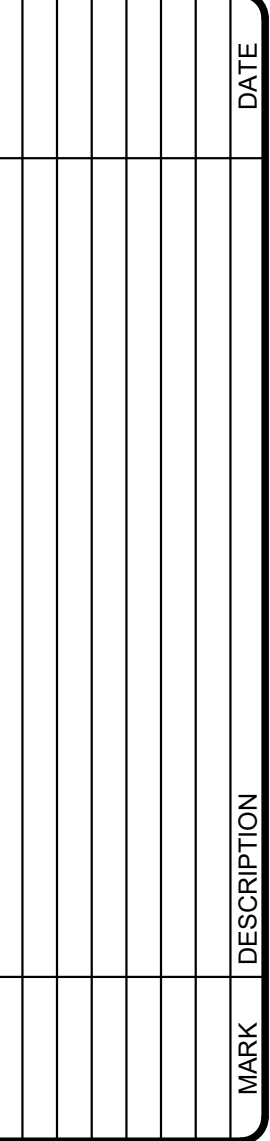
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81ST CIVIL ENGINEER SQUADRON 5081 STREET B4705 BILOXI, MS 39534		DESIGNED BY: A. TOWE		ISSUE DATE: April 26	
		DRAWN BY: M. TANNER		SOLICITATION NO.:	
		CHECKED BY:		CONTRACT NO.:	
		SUBMITTED BY: A. TOWE			
		SCALE: AS SHOWN			
		SHEET NO. 1			
		ANSI D			

KEESLER AIR FORCE BASE BILOXI, MISSISSIPPI
REPLACE MAGNOLIA DRAC AHUS, B7409
PROJECT NO.: MAHG241006

SHEET ID
M-301



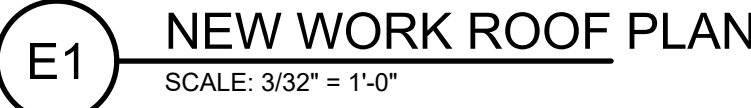


81ST CIVIL ENGINEER SQUADRON 508 L STREET B4705 BILOXI, MS 39534	SOLICITATION NO.:	
	DRAWN BY: M. TANNER	
	CHECKED BY:	CONTRACT NO.:
	SUBMITTED BY: J. CLEVELAND	
	SIZE:	ANSI D

REPLACE MAGNOLIA DFAC AHUS, B7409
PROJECT NO.: MAHG241006

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1-303



1. UFC 1 200-01: DOD BUILDING CODE.
2. UFC 3 501-01: ELECTRICAL ENGINEERING
3. UFC 3 520-01: INTERIOR ELECTRICAL SYSTEMS
4. UFC 3 550-01: EXTERIOR ELECTRICAL POWER DISTRIBUTION.
5. NFPA 70: NATIONAL ELECTRICAL CODE
6. IBC: INTERNATIONAL BUILDING CODE

SHEET ID
E-101

